

Retail Customer Intelligence Dashboard

Business Problem Statement

Background

A retail company operating across multiple product categories wants a deeper understanding of customer purchasing behavior. Leadership has observed changes in buying patterns across different customer segments, product categories, payment methods, and seasons but lacks a centralized analytics solution to explain these trends.

As a result, marketing campaigns, inventory planning, and customer retention strategies are often based on assumptions instead of data-driven insights. The company requires an interactive analytics solution to identify revenue drivers, customer preferences, and purchasing trends that can support better business decisions.

Business Question

Which customer segments, product categories, and purchasing behaviors contribute the most to business revenue, and how can these insights improve marketing effectiveness, customer retention, and inventory planning?

Business Objectives

- Identify the highest revenue-generating product categories.
- Analyze purchasing behavior across customer demographics such as age and gender.
- Evaluate the impact of subscription status, discounts, promo codes, payment methods, and shipping preferences on customer spending.
- Segment customers based on purchase frequency and spending behavior to identify high-value customers.
- Discover seasonal and regional purchasing trends.
- Build an executive dashboard that enables stakeholders to monitor key business metrics and make data-driven decisions.

Project Scope

The project covers the complete analytics lifecycle:

- **Data Preparation (Python):** Clean, validate, and transform raw customer shopping data while performing feature engineering for analysis.
- **Business Analysis (SQL):** Load the cleaned dataset into a relational database and answer business questions using joins, aggregation, Common Table Expressions (CTEs), window functions, ranking, and analytical queries.
- **Interactive Dashboard (Power BI):** Design an executive-level dashboard to monitor KPIs, customer behavior, revenue trends, and sales performance.

- **Business Report:** Summarize analytical findings and provide actionable recommendations for business stakeholders.
- **GitHub Portfolio:** Publish the complete project with documentation, source code, SQL scripts, dashboard files, screenshots, and project report.

Expected Business Outcomes

- Better understanding of customer purchasing behavior.
- Identification of high-value customer segments.
- Improved marketing and promotional strategies.
- Better inventory planning based on seasonal demand.
- Data-driven decision-making through an interactive dashboard.

Dataset

The analysis uses a retail customer shopping dataset containing 3,900 customer transactions. The dataset includes customer demographics, purchased items, product categories, purchase amount, location, season, review ratings, subscription status, shipping methods, discount usage, payment methods, purchase frequency, and other attributes required for customer behavior analysis.